

Module 2. 네트워크 구성 및 운영

2.1 장비구조와 IOS 명령어

Cisco 2600 Series Router



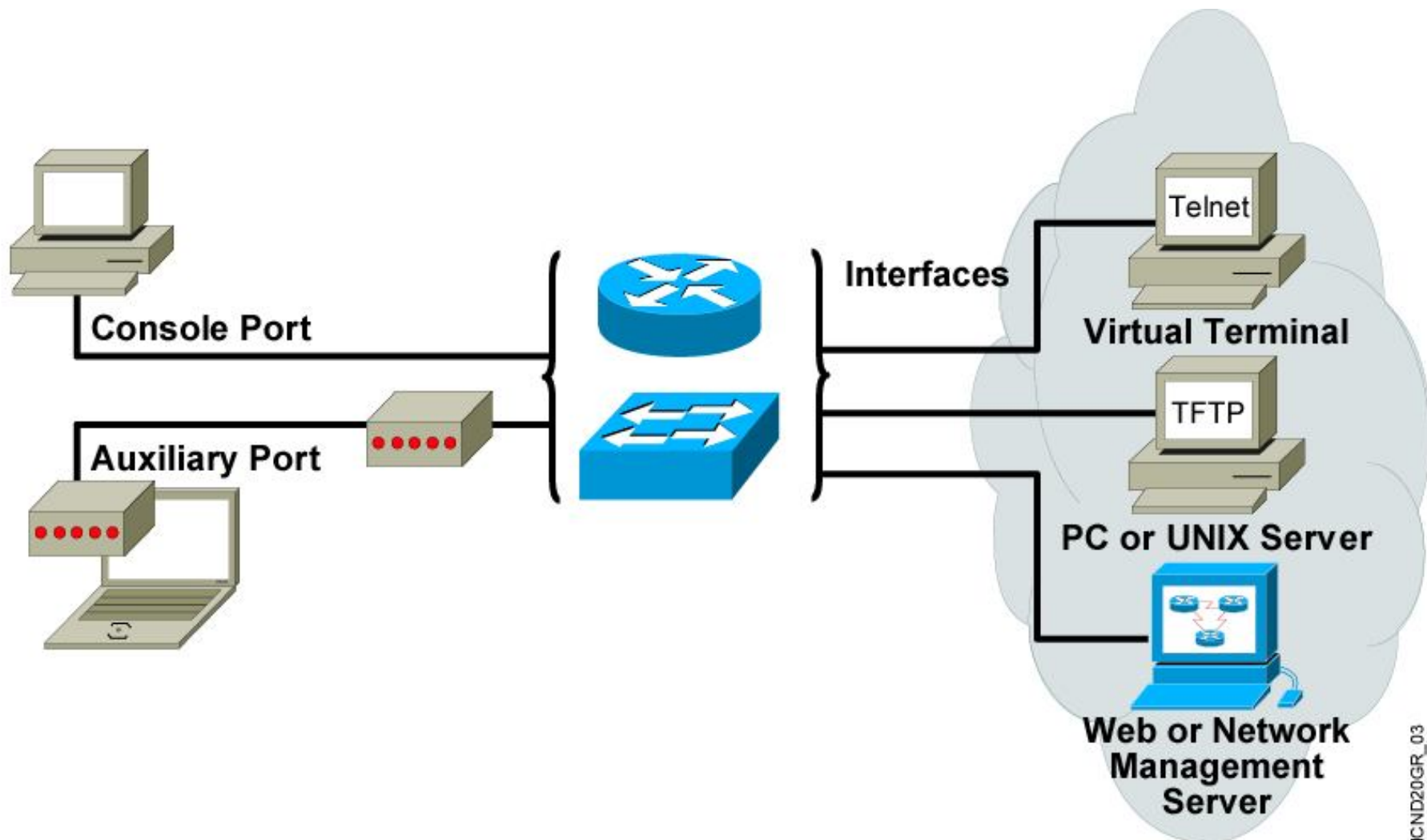
Catalyst 2950 Switch



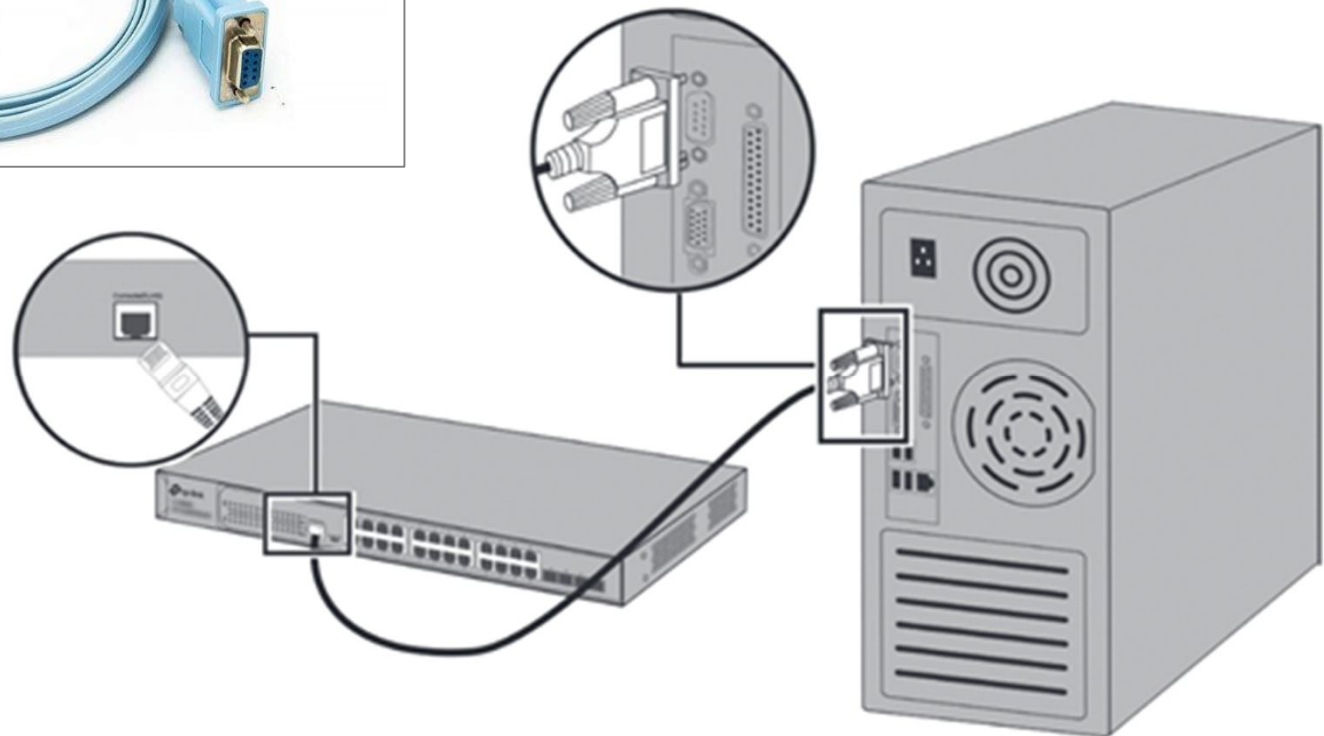
10/100Base T(RJ-45)

10/100/1000 Base T(RJ-45)

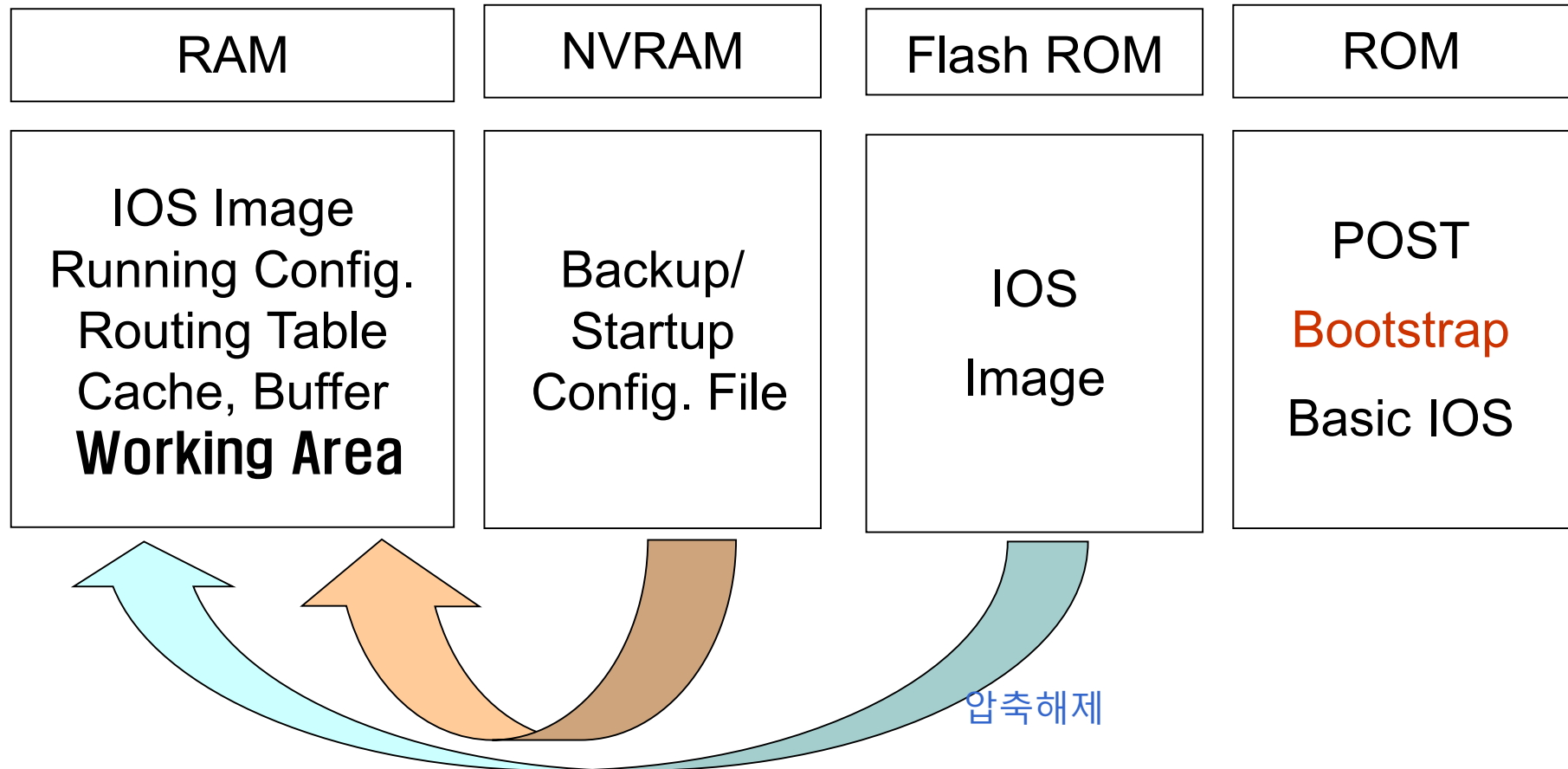
Device Manager



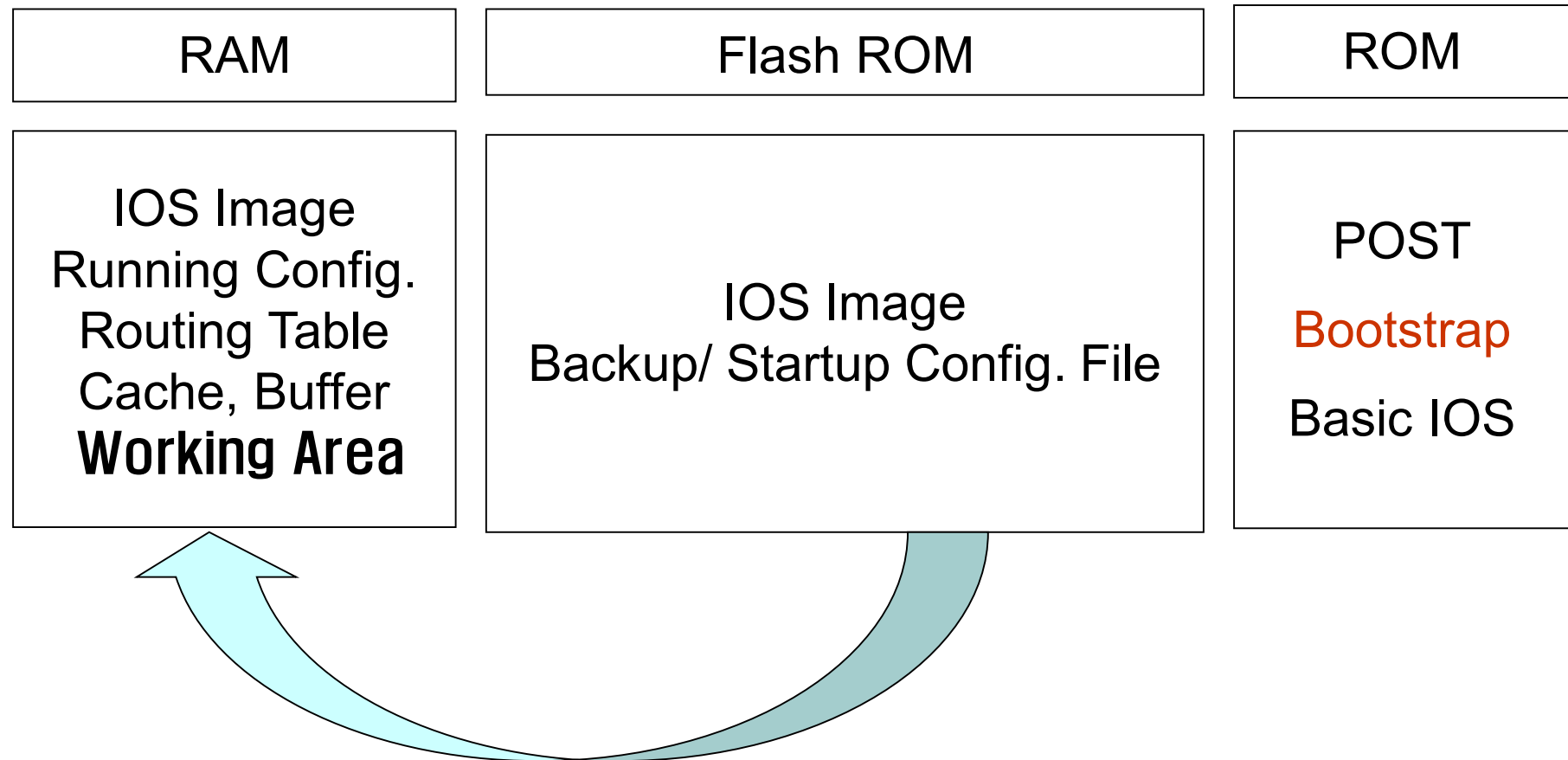
Console connection



Router Memory



Switch Memory



Router Modes

user mode
Router>

제한된 명령어만을 이용

privileged mode
Router#

모든 명령어를 이용할 수 있고
configuration file 조정할 수 있음

setup mode

초기 환경설정 privileged mode에서
명령어 setup을 입력하면 됨

RXBOOT mode

비밀번호를 잃어버렸거나 Flash의
OS가 지워지는 등의 경우 이용

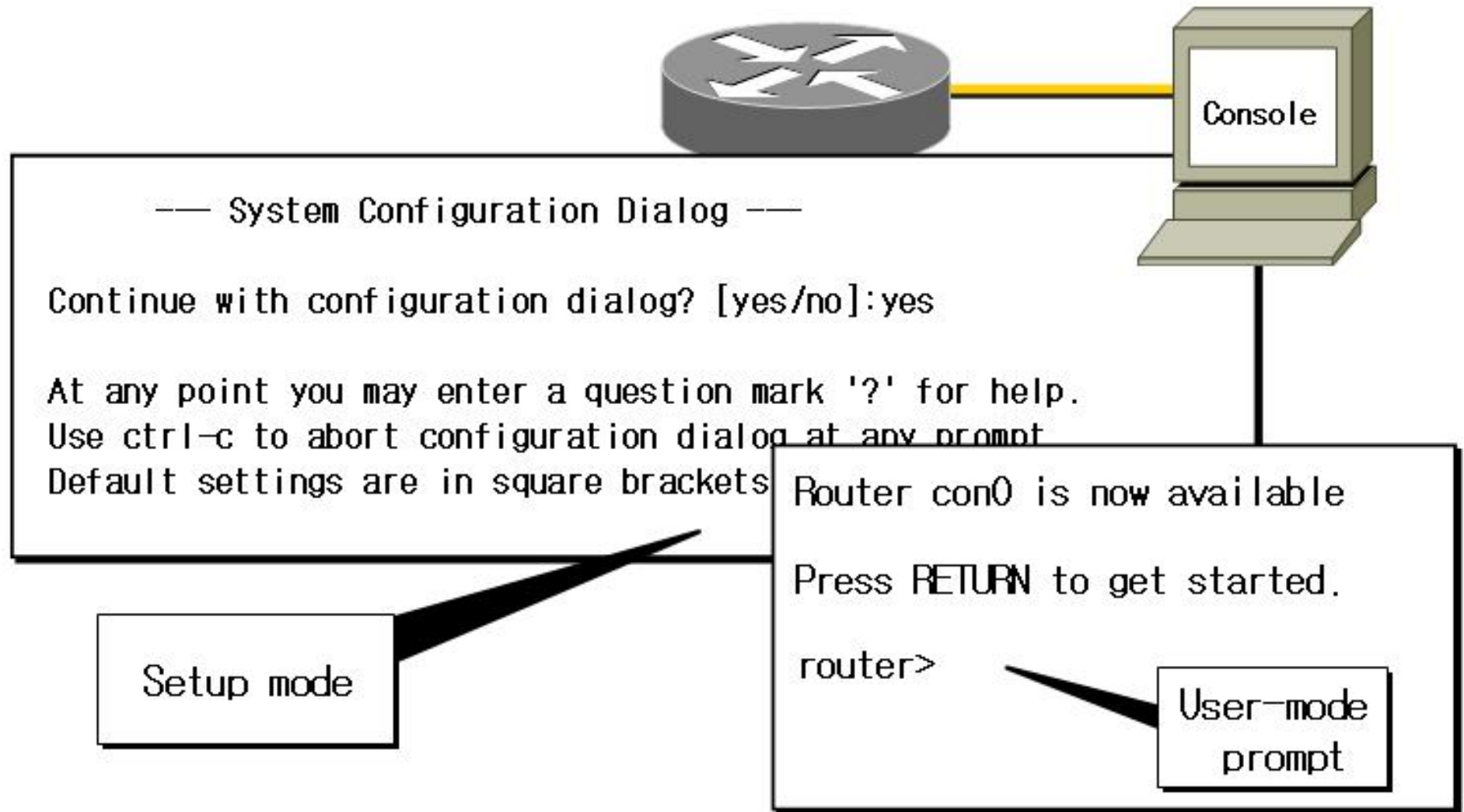
global configuration mode
Router(config)#

라우터운영 전체에 영향을 미치는
요소들을 조정할 수 있음. 하위
configuration mode로 전환할 수 있음

하위 configuration mode
Router(config-???)#

영역별 환경 설정

Setup mode configuration



Setup: The Initial Configuration Dialog

```
Router#setup
```

```
--- System Configuration Dialog ---
```

```
Continue with configuration dialog? [yes/no]: yes
```

```
At any point you may enter a question mark '?' for help.  
Use ctrl-c to abort configuration dialog at any prompt.  
Default settings are in square brackets '[]'.
```

```
Basic management setup configures only enough connectivity  
for management of the system, extended setup will ask you  
to configure each interface on the system
```

```
Would you like to enter basic management setup? [yes/no]: no
```

⊙ CLI EXEC 모드

- 명령어를 입력하기 위한 2가지 기본 모드를 제공한다.



User 모드

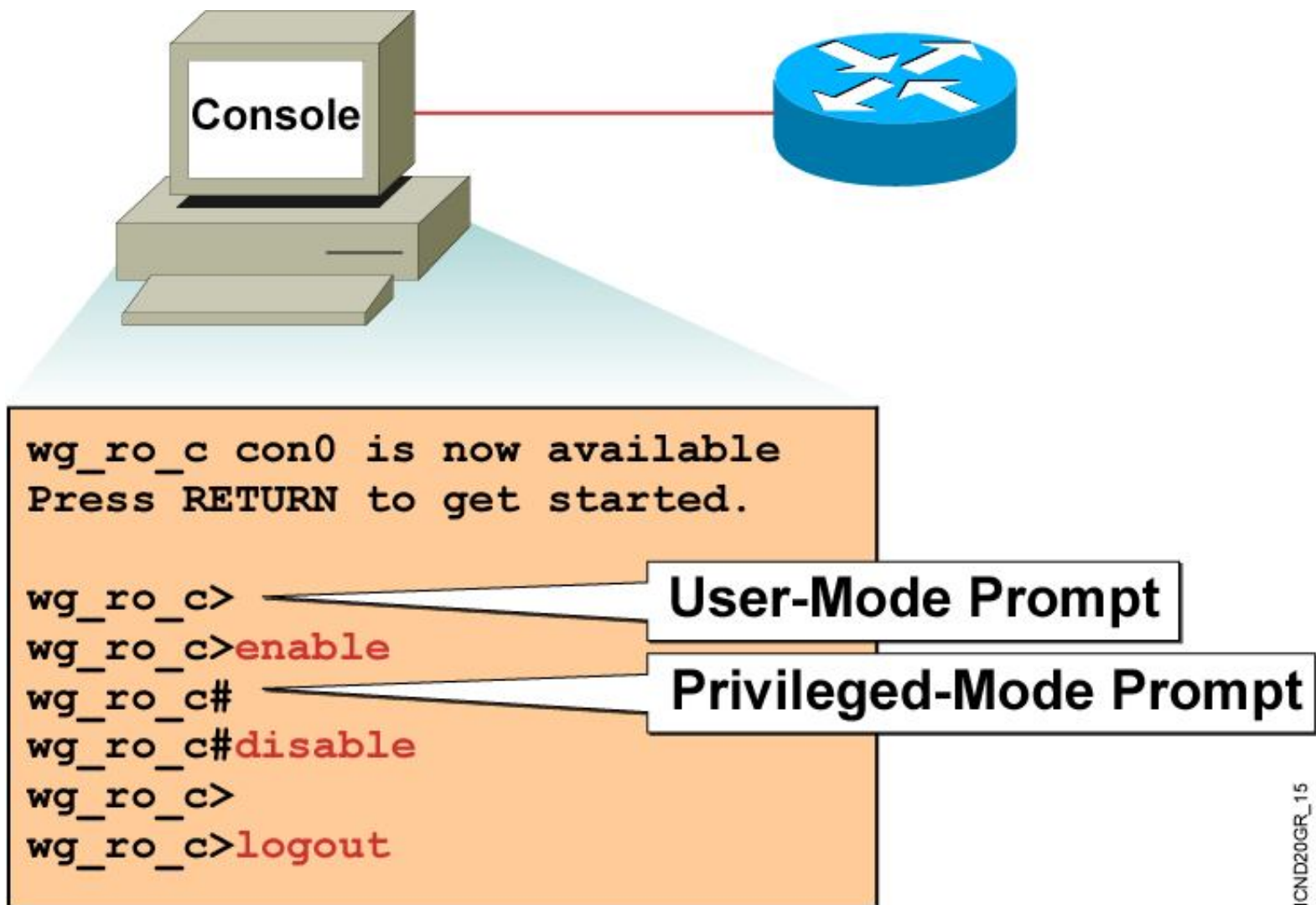
- 스위치나 라우터에 대한 제한된 검사
- 명령어 프롬프트 : `hostname >`

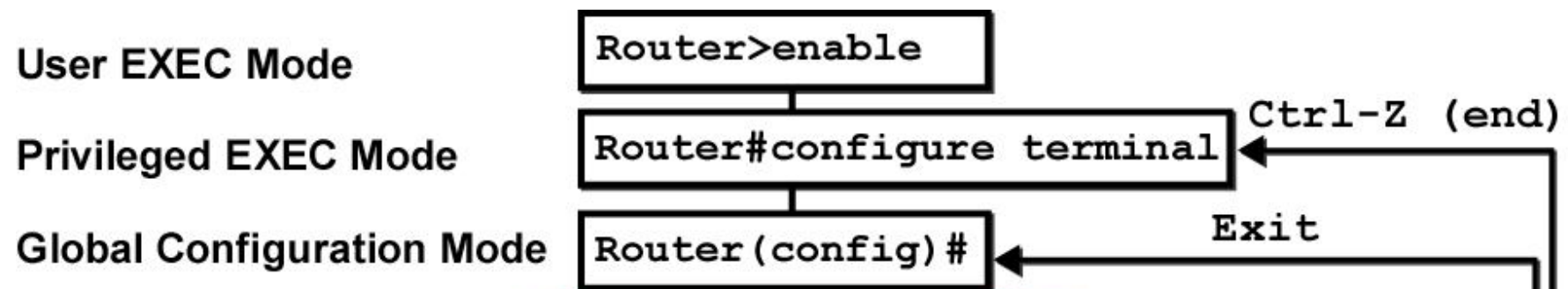
Privileged 모드

- 스위치나 라우터에 대한 자세한 검사
- configuration과 debugging 가능
- 다른 configuration 모드로 변환하기 위해 필요
- 명령어 프롬프트 : `hostname #`

- Context-Sensitive Help

```
router# cl?  
clear clock  
  
router # clock ?  
set Set the time and date  
  
router # clock set ?  
hh:mm:ss Current Time  
  
router# clock set 20:00:00 19 ?  
MONTH Month of the year  
  
router# clock set 20:00:00 19 nov ?  
<1993-2035> Year  
  
router# clock set 20:00:00 19 nov 2000  
  
router# show clock detail  
20:00:09.183 UTC Sun Nov 19 2000
```





Configuration Mode	Prompt
Interface	<code>Router(config-if)#</code>
Subinterface	<code>Router(config-subif)#</code>
Controller	<code>Router(config-controller)#</code>
Line	<code>Router(config-line)#</code>
Router	<code>Router(config-router)#</code>

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Router Name

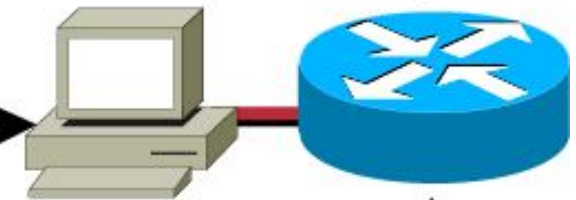
```
Router(config)#hostname wg_ro_c
wg_ro_c(config)#
```

Message-of-the-Day Banner

```
wg_ro_c(config)#banner motd #
Enter TEXT message. End with
the character #. You have
entered a secured system.
Authorized access only! #
```

Interface Description

```
wg_ro_c(config)#interface ethernet 0
wg_ro_c(config-if)#description Engineering LAN, Bldg. 18
```

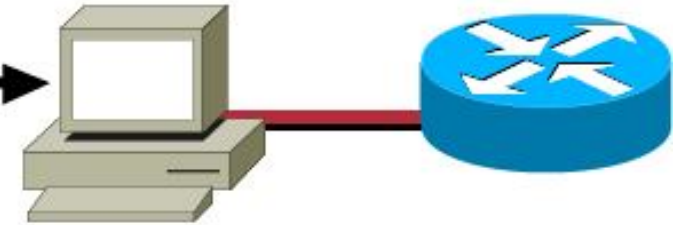


- 액세스하는 라우터 또는 인터페이스의 로컬 식별자 또는 메시지 설정

ICND20GR_18

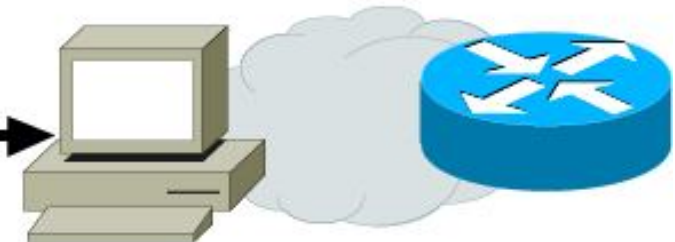
Console Password

```
Router(config)#line console 0  
Router(config-line)#login  
Router(config-line)#password cisco
```



Virtual Terminal Password

```
Router(config)#line vty 0 4  
Router(config-line)#login  
Router(config-line)#password sanjose
```



Enable Password

```
Router(config)#enable password cisco
```



Secret Password

```
Router(config)#enable secret sanfran
```

Disable Commands

```
Router(config)#no login  
Router(config)#no enable password
```

```
Router(config)#interface type number  
Router(config-if)#
```

- *type* includes serial, ethernet, token ring, fddi, hssi, loopback, dialer, null, async, atm, bri, tunnel, and so on
- *number* is used to identify individual interfaces

```
Router(config)#interface serial 0/0  
Router(config-if)#ip address 10.1.1.X 255.255.255.0  
Router(config-if)#shutdown  
Router(config-if)#no shutdown
```

- For modular routers, selects an interface

```
Router(config-if)#exit
```

- Quits from current interface configuration mode

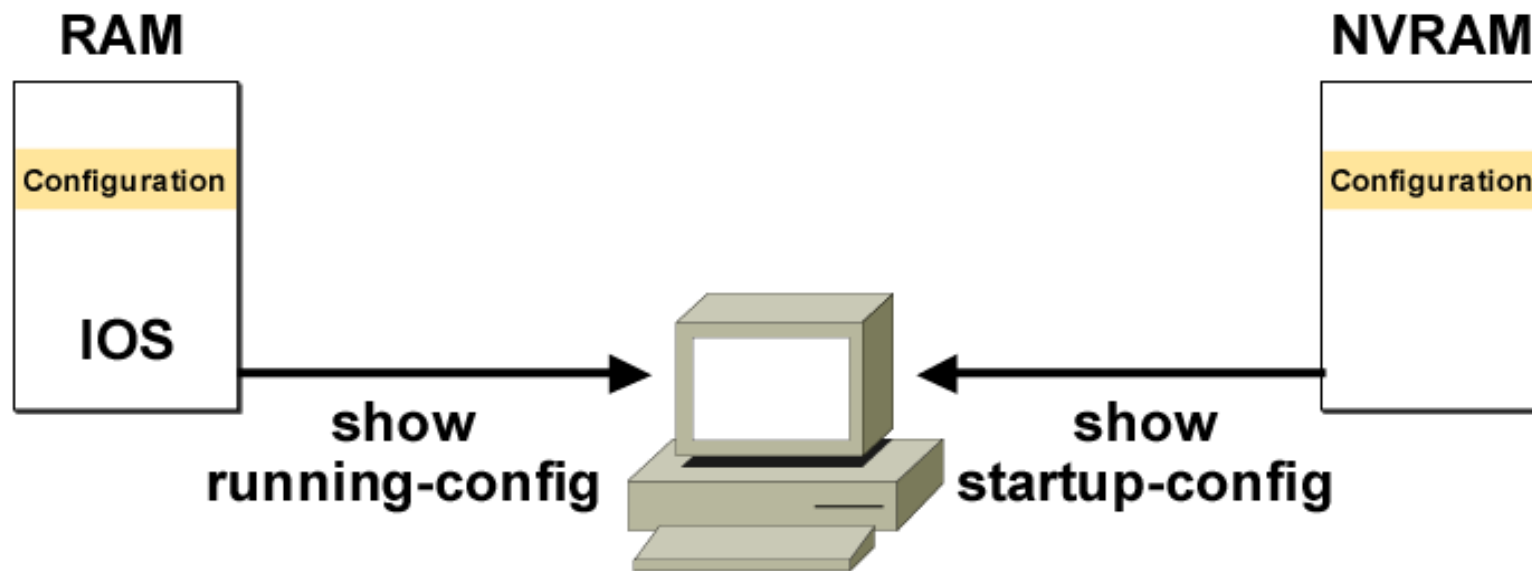
⊙ 인터페이스 disable 또는 enable

```
Router#configure terminal
Router(config)#interface serial 0
Router(config-if)#shutdown
%LINK-5-CHANGED: Interface Serial0, changed state to administratively down
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0, changed state to down
```

- Administratively turns off an interface

```
Router#configure terminal
Router(config)#interface serial 0
Router(config-if)#no shutdown
%LINK-3-UPDOWN: Interface Serial0, changed state to up
%LINEPROTO-5-UPDOWN: Line Protocol on Interface Serial0, changed state to up
```

- Enables an interface that is administratively shut down



In RAM

```
wg_ro_c#show running-config  
Building configuration...  
  
Current configuration:  
!  
version 12.0  
!  
  
-- More --
```

In NVRAM

```
wg_ro_c#show startup-config  
Using 1359 out of 32762 bytes  
!  
version 12.0  
!  
  
-- More --
```

Saving Configurations

```
wg_ro_c#  
wg_ro_c#copy running-config startup-config  
Destination filename [startup-config]?  
Building configuration...  
  
wg_ro_c#
```

- NVRAM으로 현재의 설정 정보 복사(저장)

CDP (Cisco Discovery Protocol)

- CDP는 시스코 스위치, 허브 등 Cisco IOS 10.3 또는 그 이상에서 실행된다.

- 요약된 정보에 포함되는 것들

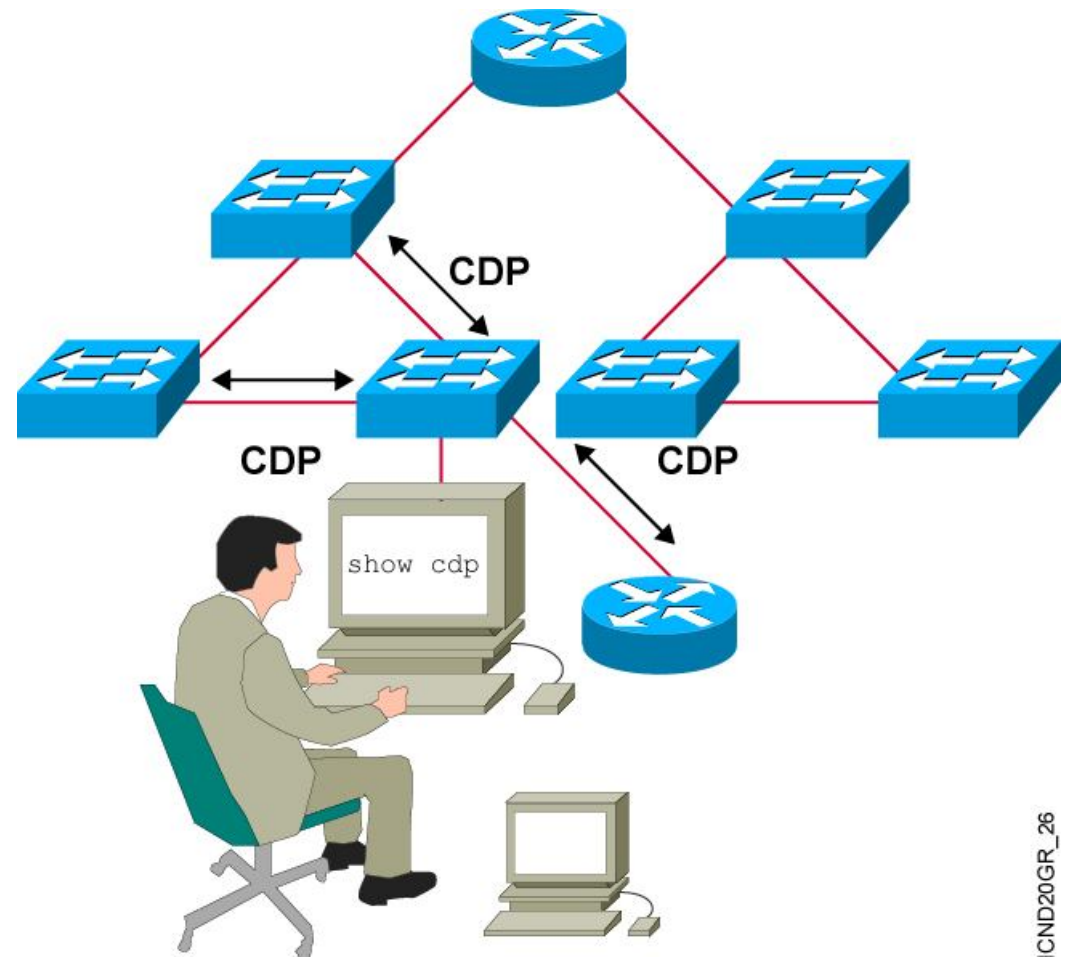
·디바이스 식별자

·주소 목록

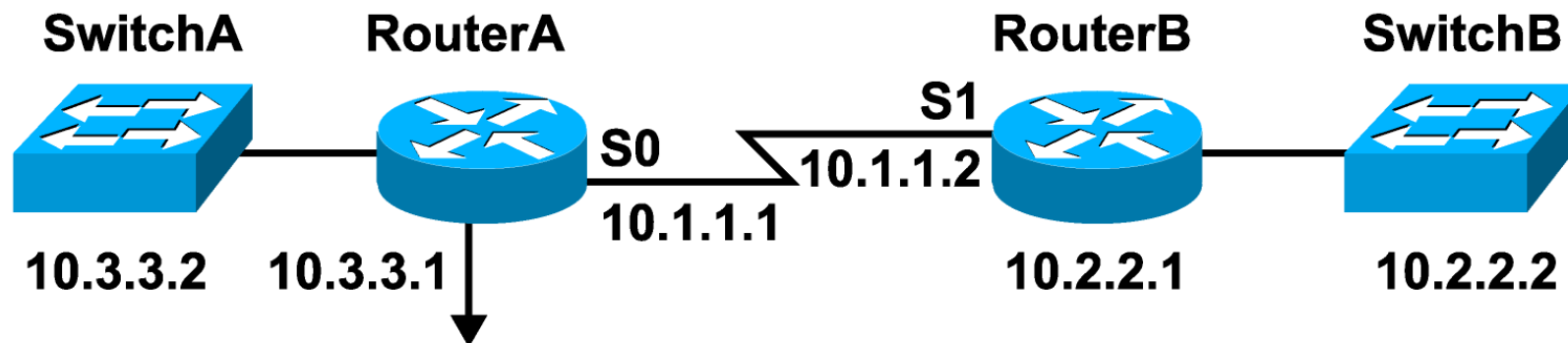
·포트 식별자

·호환성 목록

·플랫폼



CDP (Cisco Discovery Protocol)



RouterA#show cdp neighbors

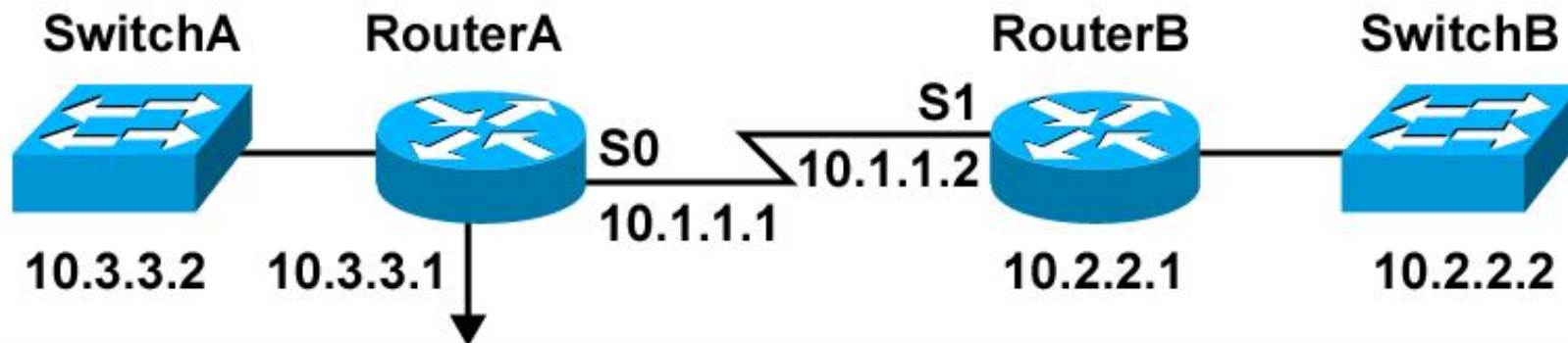
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
S - Switch, H - Host, I - IGMP, r - Repeater

Device ID	Local Intrfce	Holdtme	Capability	Platform	Port ID
RouterB	Ser 0	148	R	2522	Ser 1
SwitchA0050BD855780	Eth 0	167	T S	1900	2

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↑ SwitchA also provides its MAC address (Catalyst 1900 only).

CDP (Cisco Discovery Protocol)



```
RouterA#show cdp entry *
```

```
-----  
Device ID: RouterB
```

```
Entry address(es):
```

```
  IP address: 10.1.1.2
```

```
Platform: cisco 2522, Capabilities: Router
```

```
Interface: Serial0, Port ID (outgoing port): Serial1
```

```
Holdtime : 168 sec
```

```
Version :
```

```
Cisco Internetwork Operating System Software
```

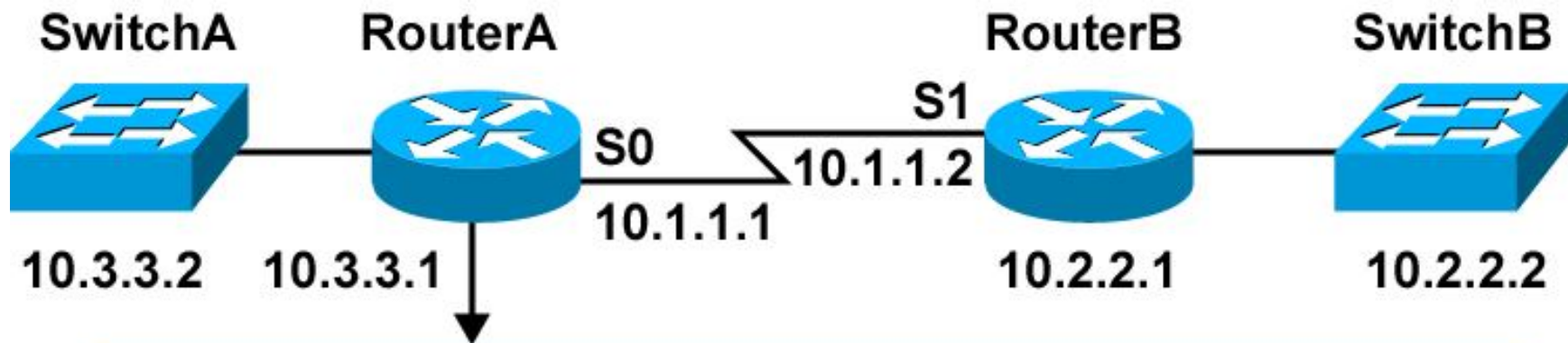
```
IOS (tm) 2500 Software (C2500-JS-L), Version 12.0(3), RELEASE SOFTWARE (fci)
```

```
Copyright (c) 1986-1999 by cisco Systems, Inc.
```

```
Compiled Mon 08-Feb-99 18:18 by phanguye
```

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CDP (Cisco Discovery Protocol)



```
RouterA#show cdp ?
  entry      Information for specific neighbor entry
  interface  CDP interface status and configuration
  neighbors  CDP neighbors entries
  traffic    CDP statistics
  <cr>
RouterA(config-if)#exit
RouterA(config)#no cdp run
RouterA(config)#interface serial0
RouterA(config-if)#no cdp enable
```

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스위치 Telnet 접속 설정

- IP 주소 설정

```
Switch(config)# int vlan1
```

```
Switch(config-if)# ip add 192.168.0.2 255.255.255.0
```

```
Switch(config-if)# no shut
```

- Gateway 설정

```
Switch(config)# ip default-gateway 192.168.0.1
```

- Telnet line 활성화

```
Switch(config)#line con 0 15
```

```
Switch(config)#login
```

```
Switch(config)# password 1234
```

라우터 SSH 접속 설정

Router# configure terminal

R1(config)# ip domain-name test.com

R1(config)# **crypto key generate rsa**

How many bits in the modulus [512]: 2048

R1(config)# username gildong privilege 15 secret 1234

R1(config)# line vty 0 4

R1(config-line)# login local

R1(config-line)# transport input ssh

R1(config-line)# exit

R1(config)#banner motd *

Authorized Access Only!

*

R1(config)#

라우터 SSH 설정 확인과 접속

show ip ssh

- **SSH Version:** 현재 활성화된 SSH 버전
 - * 보안을 위해 2.0이 권고 사항
 - R1(config)# ip ssh version 2 → SSH 버전 2(권고)
 - **Authentication Timeout:** 사용자가 접속 후 로그인을 완료해야 하는 제한 시간(초 단위)
 - **Authentication Retries:** 비밀번호를 몇 번까지 틀려도 되는지 보여주는 재시도 횟수
 - **Key Size:** 생성된 RSA 키의 길이를 확인
 - **Encryption Algorithms:** 통신에 사용되는 암호화 방식(AES, 3DES 등)을 나열
- 접속자 : ssh -l [사용자명] [라우터 IP]

라우터 초기화

```
# erase startup-config
```

```
# reload
```

Switch 초기화

erase start → config.txt 파일 삭제 (router의 startup-config와 동일)

delete vlan.dat → vlan.dat 파일 삭제

reload

sh flash → vlan.dat 파일과 config.txt 파일 확인 가능

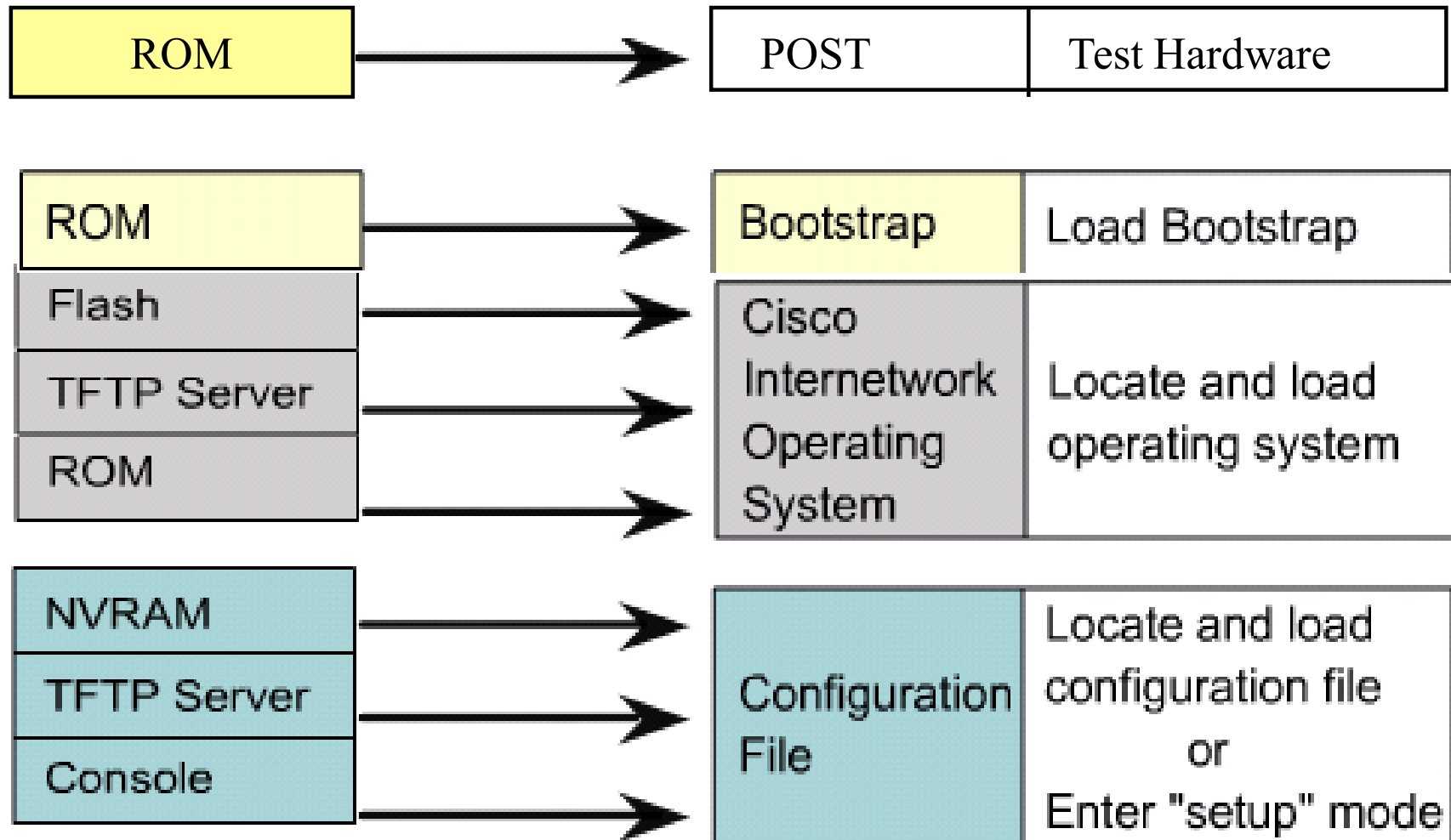
sh mac → MAC table 확인 가능

- 스위치는 Flash에 IOS, config 파일을 동시에 갖고 있음
- NVRAM이 존재하지 않음

장비 부팅 순서

- POST (Power On Self Test)
- 부트 스트랩코드 적재와 실행
- 시스코 IOS 소프트웨어 찾기
- 시스코 IOS 소프트웨어 적재(메모리에 load) 설정 정보 찾기
 - 만약 NVRAM 안에 설정 내용이 없다면 setup 모드로 들어간다.
- 설정 정보 적재(start config 메모리에 load)
- 설정 정보로 시스코 IOS 소프트웨어 실행

장비 부팅 순서



Configuration Register Value

```
wg_ro_a#show version
Cisco Internetwork Operating System Software
IOS (tm) 2500 Software (C2500-JS-L), Version 12.0(3), RELEASE SOFTWARE (fc1)
Copyright (c) 1986-1999 by cisco Systems, Inc.
Compiled Mon 08-Feb-99 18:18 by phanguye
Image text-base: 0x03050C84, data-base: 0x00001000

ROM: System Bootstrap, Version 11.0(10c), SOFTWARE
BOOTFLASH: 3000 Bootstrap Software (IGS-BOOT-R), Version 11.0(10c), RELEASE SOFTWARE (fc1)

wg_ro_a uptime is 20 minutes
System restarted by reload
System image file is "flash:c2500-js-l_120-3.bin"

--More--

Configuration register is 0x2102
```

Locating Cisco IOS Software

- ❑ Configuration Register 0X2102 → 정상 boot
 - NV-RAM Boot system 명령 → Flash → TFTP → ROM
 - LAB_A> enable
Password: test
 - LAB_A#
- ❑ Configuration Register 0X2100 → ROM 상에서 Booting image호출
Rommon 1 > → ROM monitor
- ❑ Configuration Register 0X2101 → MiniIOS에 있는 Boot-image 이용
 - Rom(boot) >
- ❑ Configuration Register 0X2142 → Startup-config 무시, 정상 boot
 - Dialog setup? No or Ctrl+C
 - Router> enable
 - Router#

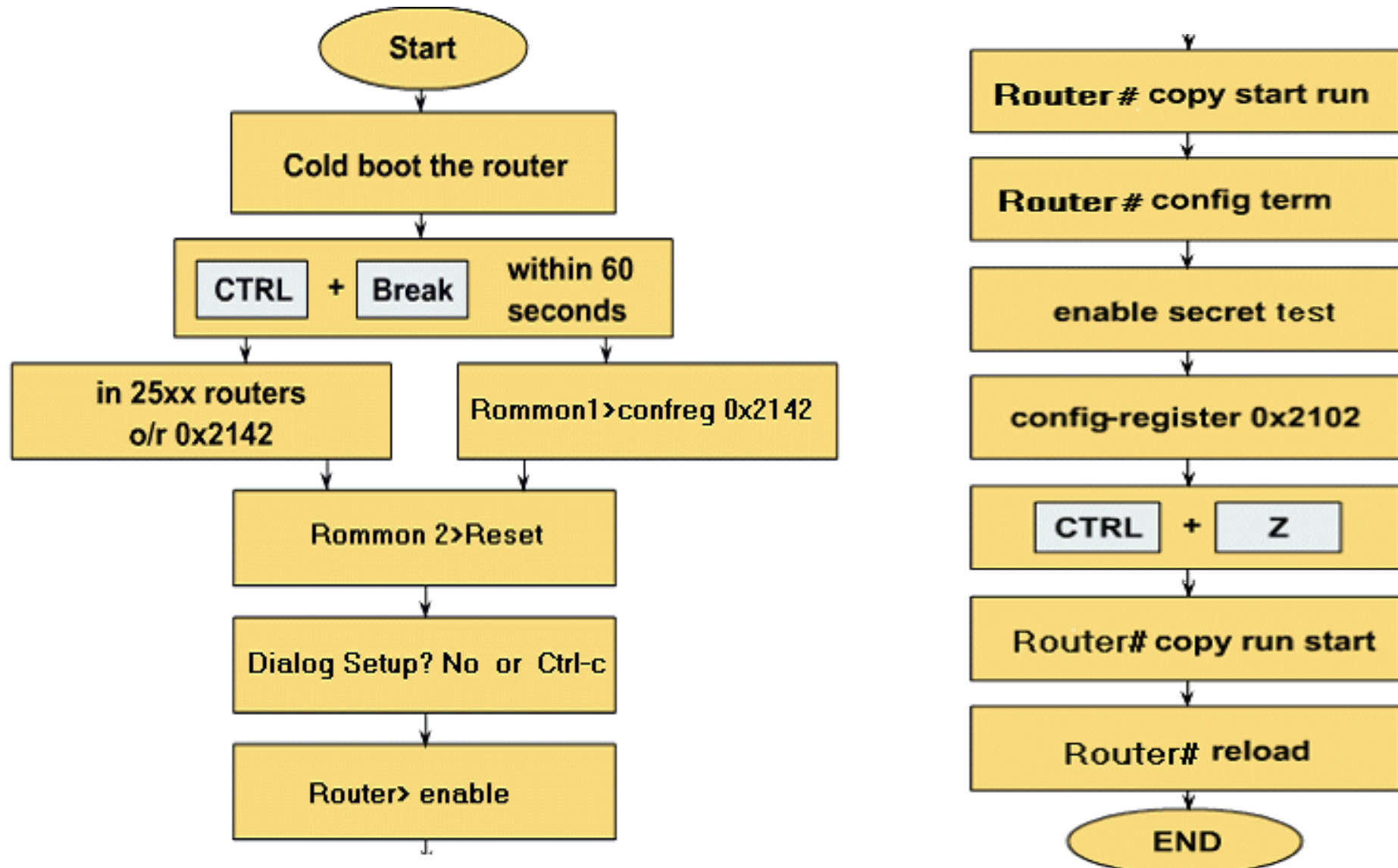
Configuration Register Values

Value	Description
0xnnn0	Use ROM monitor mode (<u>manually boot using the b command</u>)
0xnnn1	Automatically boot from ROM (default if router has no Flash)
0xnnn2 to 0xnnnF	Examine NVRAM for boot system commands (0xnnn2 is the default if the router has Flash)

■ Configuration register의 Boot Field setting

- router(config)# config-register 0x2102
- rommon1> confreg 0x2102
- NVRAM(startup-config)의 boot system 명령에 따름
- default : Flash
- ex) (0x 2100) – rommon (0x 2101) – rom(boot)
(0x 2102) – flash,startup (0x 2142) – flash,startup(x)

Router Password Recovery



TFTP (Trivial FTP)

- TFTP 서버에 config 파일 백업

copy run tftp

IP ? TFTP 서버의 IP 주소 입력

파일명 ? Default

!!!!!! 정상 작동

- TFTP 서버에 IOS 파일 백업

sh flash ➔ 명령어를 통해 확인 ex) c2500-jk80...

copy flash tftp

Source filename ? IOS 파일명 붙여 넣기

? TFTP IP 주소 입력

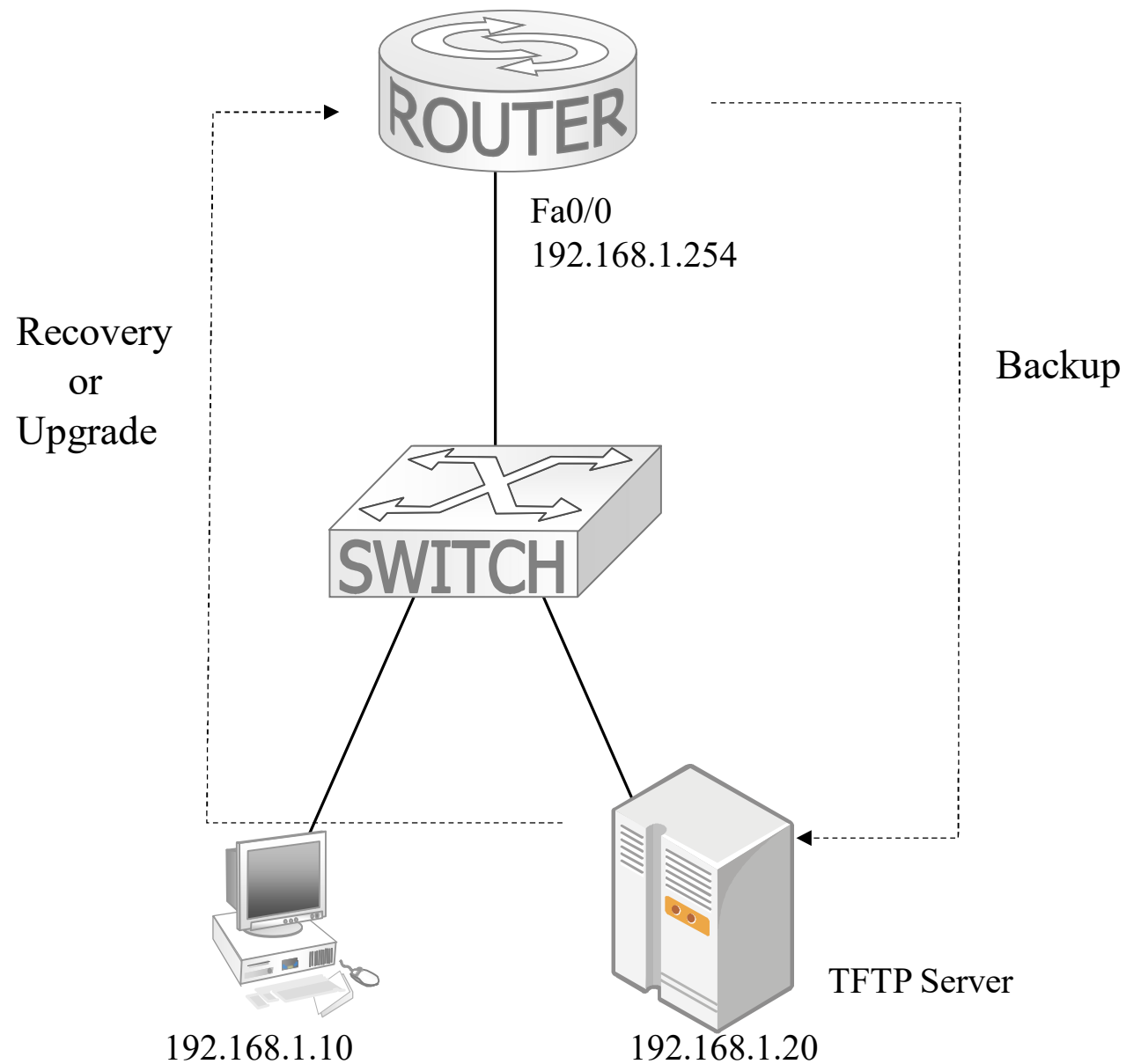
TFTP를 이용한 Configuration Files 관리

```
GAD#copy running-config tftp
Address or name of remote host
[]?192.168.119.20
Destination filename [GAD-config]?
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
624 bytes copied in 7.05 secs
GAD#copy tftp running-config
Address or name of remote host []?
192.168.119.20
Source filename []?GAD-config
Destination filename [running-config]?
Accessing tftp://192.168.119.20/GAD-
config...
Loading GAD-config from 192.168.119.20
(via FastEthernet 0/0):
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
[OK-624 bytes]
624 bytes copied in 9.45 secs
GAD#
```

TFTP를 이용한 IOS Image Upgrade

```
GAD#copy tftp flash
Address or name of remote host []?192.168.119.20
Source filename []? C2600-js-l_121-3.bin
Destination filename [C2600-js-l_121-3.bin]?
Accessing tftp://192.168.119.20/ C2600-js-l_121-3.bin
Erase flash: before copying? [confirm]
Erasing the flash file system will remove all files
Continue? [confirm]
Erasing device eeeeeee.....erased
Loading C2600-js-l_121-3.bin from 192.168.119.20 (via
FastEthernet 0/0): !!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
Verifying Check sum .....OK
[OK-8906589 bytes]
8906589 bytes copied in 277.45 secs
GAD#
```

IOS image Backup/Upgrade



본 교재는 수강생 개인 학습용으로만 제공됩니다.

무단 복제, 배포, 수정, 공유, 온라인 게시 행위를 금합니다.

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